

Physic

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1 → (3)

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6 → The ratio of speed of light in vacume and speed of light in that medium is known as reflective index.

speed of light in absolut medium = $1.5 \times 10^8 \text{ m/s}$
 " " " " vacume = $3 \times 10^8 \text{ m/s}$

speed of light = reflective index in vacume

reflective index in absolut transparent medium.

$$= \frac{3 \times 10^8}{1.5 \times 10^8} = 2 \text{ m/s.}$$

7. The fuse chances of appliance pass throu circuit in fuse a large circuit circuit

8. normal correct

$$\frac{1}{f} =$$

$$\Rightarrow \frac{1}{f} =$$

$$\Rightarrow \frac{1}{f} =$$

$$\Rightarrow f =$$

$$\Rightarrow P$$

$$= P$$

7. The fuse used in the series to reduce the chances of burning of any electrical appliance when large amount of electricity pass through the it, it ~~break~~ break the circuit when ~~joule~~ joule heating takes place. ~~the~~ fuse melt. If it replaced by the a large rating then fuse don't melt and circuit continued and due to continued circuit application catch fire.

8. normal vision = $u = -25\text{cm}$
corrected " = $v = -75\text{cm}$

$$\frac{1}{P} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{P} = \frac{1}{-75} - \frac{1}{-25}$$

$$\Rightarrow \frac{1}{P} = \frac{-1 + 3}{-75} = \frac{2}{75} \Rightarrow$$

$$\Rightarrow P = \frac{75}{2} = 37.5\text{ cm} = 0.375\text{ m}$$

$$\Rightarrow P = \frac{1}{f} = \frac{1}{0.375} = \frac{1000}{375}$$

$$= P = \frac{8}{3} = 2.67\text{ D}$$

defect is hypermetropia.
 Power = 2.67 D
 focal length = 37.5 cm

9. $R = \frac{\rho l}{A}$

given, $l = 2l$
 $A = \frac{A}{2}$
 $\rho = \text{constant}$

(ii) $R = \frac{\rho \cdot 2l}{A/2}$
 $= \frac{\rho \cdot 2l \times 2}{A}$
 $= \rho \cdot \frac{4l}{A} \quad \text{or} = 4 \frac{\rho l}{A}$
 $R_n = 4R$

(ii) resistivity remains same because it does not depend on length and cross-section.

10. (i) P puts you into the plane of paper, and Q takes you out of it. The strength of the magnetic field is larger at the ^{point} ~~source~~ which is closer to ~~magnetic~~ source, which is Q .



(ii) a) By $\dots$ shows
 b) The dis
 center
 perpend

11.

(i) In

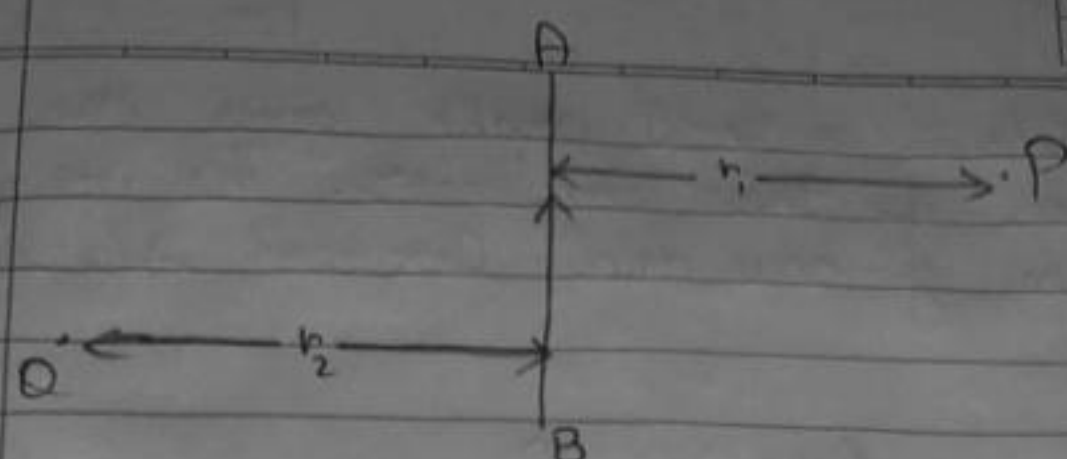
R

V

In

P

H



(ii) a) By using Compauss. method, whose middle shows the direction.

b) The direction of magnetic field at the center of a current-carrying loop is perpendicular to the plane of the loop.

11.

(i) In case of $10 \Omega R_1$

$$R_1 = 10 \Omega$$

$$V = 3$$

$$\therefore \text{then, } g = \frac{V}{R} = \frac{3}{10}$$

$$g = 0.3$$

In case of $15 \Omega R_2$

$$R_2 = 15 \Omega$$

$$V = 3$$

$$\text{then, } g = \frac{V}{R} = \frac{3}{15} = \frac{2}{5}$$

$$= 0.2 \Omega$$